

ECO 362 Financial Investments

Precept 3: Risk and Return, Diversification Primer

September 22 – 25, 2025

1 Cumulative Return

2 Portfolio Choice

3 Rebalancing

4 Diversification

- An **asset i** pays a stream of random payoffs $X_{i,t}$, $t = 1, \dots, +\infty$
- Sell the asset at price $P_{i,t}$ at period t , after collecting the payoff $X_{i,t}$
- What is the **return of asset i** at period t ?
 - " the rate of return if you purchase the asset at period $t - 1$ and sell it at period t "

$$r_{i,t} = \frac{(P_{i,t} + X_{i,t})}{P_{i,t-1}} - 1$$

- Calculate cumulative return from monthly return:

$$1 + r_{cumulative,T} = \prod_{t=1}^T (1 + r_{i,t})$$

- For portfolio, calculate the portfolio monthly return first:

$$r_{p,t} = \sum_i w_{i,t-1} r_{i,t}.$$

Summation is over all assets in the portfolio.

- Example: two assets: risk-free bond and stock, with weight $1 - w_{t-1}$ and w_{t-1}

$$r_{p,t} = (1 - w_{t-1})r_{b,t} + w_{t-1}r_{s,t}.$$

"Risk Examples" Example 1: Cumulative Returns

- Comparison of cumulative returns of stocks and bonds
- Mean and standard deviation

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Tradeoff between Risk and Return

- Let $r_{b,t}$ and $r_{s,t}$ be the returns of bond and stock respectively.
- A portfolio with proportion w of wealth in the stock has return:

$$r_{p,t} = (1 - w)r_{b,t} + w \cdot r_{s,t}$$

- Expected return:

$$\mathbb{E}[r_{p,t}] = r_{b,t} + w(\mathbb{E}[r_{s,t}] - r_{b,t})$$

- Return of bond is risk-free: no volatility in $r_{b,t}$. Portfolio volatility:

$$\sigma_{p,t} = w \cdot \sigma_{s,t}$$

- Capital market line (CML):

$$\mathbb{E}[r_{p,t}] = r_{b,t} + \frac{\mathbb{E}[r_{s,t}] - r_{b,t}}{\sigma_{s,t}} \sigma_{p,t}$$

- Quadratic utility ($\gamma \geq 0$):

$$U(1 + r_p) = 1 + r_p - \frac{\gamma}{2}(r_p - \mathbb{E}[r_p])^2$$

- In expectation, this gives rise to mean-variance expected utility:

$$\mathbb{E}[U(1 + r_p)] = 1 + \mathbb{E}[r_p] - \frac{\gamma}{2}\sigma_p^2$$

- Choose portfolio w to maximize expected utility:

$$\mathbb{E}[U] = r_b + w(\mathbb{E}[r_s] - r_b) - \frac{\gamma}{2} w^2 \sigma_s^2$$

First-order condition with respect to portfolio weight:

$$\frac{d\mathbb{E}[U]}{dw} = \mathbb{E}[r_s] - r_b - \gamma w \sigma_s^2 = 0$$

Solve for optimal portfolio weight:

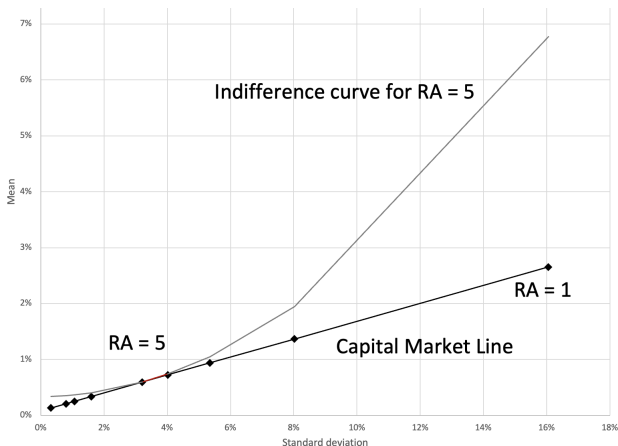
$$w = \frac{\mathbb{E}[r_s] - r_b}{\gamma \sigma_s^2}$$

- Set of points $(\sigma_p, \mathbb{E}[r_p])$ that give the same expected utility
$$\bar{U} := \mathbb{E}[U(1 + r_p)] = 1 + \mathbb{E}[r_p] - \frac{\gamma}{2}\sigma_p^2$$
- Certainty equivalent: riskless rate of return that gives you same expected utility: $\mathbb{E}[U(1 + r_{CE})] = \bar{U} \implies 1 + r_{CE} = \bar{U}$.
- Indifference curve is given by

$$\mathbb{E}[r_p] = r_{CE} + \frac{\gamma}{2}\sigma_p^2$$

"Risk Examples" Example 2: Mean-Variance Portfolio Choice

- Compute optimal portfolio weights for different value of risk aversion γ
- Plot capital market line (CML): $\mathbb{E}[r_{p,t}] = r_{b,t} + \frac{\mathbb{E}[r_{s,t}] - r_{b,t}}{\sigma_{s,t}} \sigma_{p,t}$
- Draw indifference curve ($\mathbb{E}[r_p] = r_{CE} + \frac{\gamma}{2} \sigma_p^2$) for $\gamma = 5$



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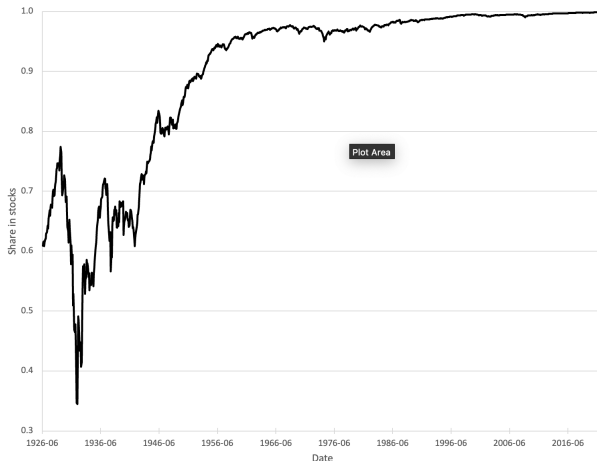
- Portfolio weight changes after each period's return is realized

$$w_t = \frac{w_{t-1}(1 + r_{s,t})}{(1 - w_{t-1})(1 + r_{b,t}) + w_{t-1}(1 + r_{s,t})}$$

- If bond return is higher than the stock return ($r_{b,t} > r_{s,t}$), then the share of bond holdings increases, or the share of stock holdings decreases ($w_t < w_{t-1}$).
- If your risk aversion and the mean/variance of stock returns do not change, then theory prescribes a constant portfolio weight w .
- Hence, in the situation above, you need to **sell bonds** and **buy stocks** to restore the optimal portfolio weight.

"Risk Examples" Example 3: Importance of Rebalancing

- Example 3 computes portfolio weights over time without rebalancing.



- In the data, stocks have higher return than bonds in the long run, and thus stocks will dominate the portfolio if you don't rebalance.

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- Assume that the return on stock i is

$$r_i = r_m + \epsilon_i$$

- r_m is the market return, which represents the *systematic* risk
- ϵ_i is the *idiosyncratic* risk of stock i
- ϵ_i has mean 0 and variance σ_ϵ^2 and ϵ_i is uncorrelated with r_m and ϵ_j for $j \neq i$, that is,

$$\text{Cov}(\epsilon_i, \epsilon_j) = \text{Cov}(\epsilon_i, r_m) = 0$$

Benefits of Diversification under the market model

- Consider an equal-weighted portfolio

$$r_p = \frac{1}{I} \sum_{i=1}^I r_i = r_m + \frac{1}{I} \sum_{i=1}^I \epsilon_i$$

- The mean is $\mathbb{E}(r_p) = \mathbb{E}(r_m)$
- The variance is

$$\begin{aligned} \text{Var}(r_p) &= \text{Var}(r_m) + \frac{2}{I} \sum_{i=1}^I \text{Cov}(r_m, \epsilon_i) + \frac{1}{I^2} \sum_{i=1}^I \sum_{j=1}^I \text{Cov}(\epsilon_i, \epsilon_j) \\ &= \text{Var}(r_m) + \frac{1}{I^2} \sum_{i=1}^I \text{Cov}(\epsilon_i, \epsilon_i) \\ &= \text{Var}(r_m) + \frac{1}{I} \sigma_\epsilon^2 \rightarrow \text{Var}(r_m) \text{ when } I \rightarrow +\infty \end{aligned}$$